KAELEARN // LINEAR ALGEBRA I // MC PREP

Tick or Trap

All 27 pool questions as a 100-box drill, the traps behind the distractors, and the key.



HOW TO USE IT // 25 MINUTES

01 Part I cold: no notes, no Musterantworten, 25 minutes. Tick every true statement. Any number of boxes per question can be true, from none to all four.
02 Score with the key in Part III: one point per box, for a tick on a true statement or a blank on a false one. 100 points.
03 Every miss: write the true statement in the margin, then find its trap in Part II.
04 Two days before the test, Part I again, cold. Only the boxes you missed matter.

PART I

The Drill

01 Let $f\colon \mathbb{R} \to \mathbb{R}, x \mapsto x^2 + 1$.

- ☐ *a* $f(-1) = 0$.
- ☐ *b* The image *(Bild)* of f is $[1, \infty)$.
- ☐ *c* f is surjective.
- ☐ *d* f is injective.

02 Vectors $v_1, \dots, v_k$ in a real vector space are linearly independent *(linear unabhängig)* if and only if

- ☐ *a* $\lambda_1 v_1 + \cdots + \lambda_k v_k = 0$ holds only for $\lambda_1 = \cdots = \lambda_k = 0$.
- ☐ *b* $\lambda_1 v_1 + \cdots + \lambda_k v_k = 0$ holds for $\lambda_1 = \cdots = \lambda_k = 0$.
- ☐ *c* every v_i is nonzero.
- ☐ *d* (for $k \geq 2$) no v_i is a linear combination of the others.

03 Let G be a set with an operation $*\colon G \times G \to G$.

- ☐ *a* $(G, *)$ is a group *(Gruppe)* if $*$ is associative, has a neutral element, and every element has an inverse.
- ☐ *b* In every group, $a * b = b * a$ for all a, b.
- ☐ *c* $(\mathbb{N}, +)$ is a group.
- ☐ *d* $(\mathbb{R} \setminus \{0\}, \cdot)$ is a group.

04 Lines *(Geraden)* in $\mathbb{R}^n$, $n \geq 2$.

- ☐ *a* Two points $p \neq q$ determine exactly one line, $\{p + t(q - p) \mid t \in \mathbb{R}\}$.
- ☐ *b* Any two points determine a line.
- ☐ *c* A line through the origin is $\operatorname{span}(v)$ for a single vector $v \neq 0$.
- ☐ *d* $\{t(1,2) \mid t \in \mathbb{R}\}$ and $\{t(2,4) \mid t \in \mathbb{R}\}$ are different lines.

05 Let $A = \{1,2,3\}$ and $B = \{3,4\}$.

- ☐ *a* $A \cup B = \{1,2,3,4\}$
- ☐ *b* $A \cap B = \{3\}$
- ☐ *c* $3 \notin A \cup B$, as "or" excludes "both"
- ☐ *d* $A \cap \emptyset = A$

06 Let $z = 1 + 2i$ and $w = 1 - i$.

- ☐ *a* $z + w = 2 + i$
- ☐ *b* $zw = -1 + i$
- ☐ *c* $z/w = (-1 + 3i)/2$
- ☐ *d* $i^2 = 1$

07 Which are real vector spaces *(reelle Vektorräume)*, with the usual addition and scalar multiplication?

- ☐ *a* $\mathbb{R}^3$
- ☐ *b* $P_2(\mathbb{R})$, the real polynomials of degree at most 2
- ☐ *c* the real polynomials of degree exactly 2
- ☐ *d* $\{(x,y) \in \mathbb{R}^2 \mid x + y = 1\}$

08 Which maps are injective but not surjective?

- ☐ *a* $\mathbb{R} \to \mathbb{R}, x \mapsto e^x$
- ☐ *b* $\mathbb{R} \to (0,\infty), x \mapsto e^x$
- ☐ *c* $\mathbb{N} \to \mathbb{N}, n \mapsto n + 1$
- ☐ *d* $\mathbb{R} \to \mathbb{R}, x \mapsto x^3$

09 Which sets are linearly dependent over $\mathbb{R}$? Here $e_1 = (1, 0, \ldots, 0) \in \mathbb{R}^8$.

- ☐ *a* $\{(1,2),(2,4)\} \subset \mathbb{R}^2$
- ☐ *b* $\{1,2\} \subset \mathbb{R}$
- ☐ *c* $\{e_1, 2e_1\} \subset \mathbb{R}^8$
- ☐ *d* $\{(1,1),(1,-1)\} \subset \mathbb{R}^2$

10 Fields *(Körper)*.

- ☐ *a* In a field, every element has a multiplicative inverse.
- ☐ *b* $\mathbb{R}$ and $\mathbb{C}$, with $+$ and $\cdot$, are fields.
- ☐ *c* $\mathbb{Q}$ is a field.
- ☐ *d* In every field, $0 \neq 1$.

11 Planes and hyperplanes *(Ebenen und Hyperebenen)*.

- ☐ *a* A hyperplane in $\mathbb{R}^n$ has dimension $n - 1$.
- ☐ *b* In $\mathbb{R}^3$, the hyperplanes are exactly the planes.
- ☐ *c* In $\mathbb{R}^2$, the hyperplanes are exactly the lines.
- ☐ *d* In $\mathbb{R}^4$, every plane is a hyperplane.

12 Let $f\colon X \to Y$ be a map.

- ☐ *a* f is injective $\iff$ for all $x_1, x_2 \in X$: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.
- ☐ *b* f is injective $\iff$ for all $x_1, x_2 \in X$: $x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$.
- ☐ *c* f is surjective $\iff$ $f(X) = Y$.
- ☐ *d* f is bijective $\iff$ every $y \in Y$ has exactly one preimage *(Urbild)*.

13 The Euclidean norm *(Norm)* on $\mathbb{R}^n$.

- ☐ *a* $\|(3,4)\|_2 = 5$.
- ☐ *b* $\|\cdot\|_2$ is a map $\mathbb{R}^n \to \mathbb{R}_{\geq 0}$: every x gets exactly one number ≥ 0.
- ☐ *c* $\|x+y\|_2 \leq \|x\|_2 + \|y\|_2$ for all $x, y \in \mathbb{R}^n$.
- ☐ *d* $\|\lambda x\|_2 = \lambda \|x\|_2$ for all $\lambda \in \mathbb{R}$, $x \in \mathbb{R}^n$.

14 Two or three vectors.

- ☐ *a* Two vectors in $\mathbb{R}$ (over $\mathbb{R}$) can be linearly independent.
- ☐ *b* Two vectors in $\mathbb{R}^2$ can be linearly independent.
- ☐ *c* Three vectors in $\mathbb{R}^2$ can be linearly independent.
- ☐ *d* $u, v \in \mathbb{R}^n$ are linearly dependent $\iff$ one of them is a multiple of the other.

15 Maps $f\colon A \to B$, $g\colon B \to C$ and $h\colon A \to C$ form a triangle.

- ☐ *a* The triangle is a commutative diagram *(kommutatives Diagramm)* $\iff h = g \circ f$.
- ☐ *b* $g \circ f$ means: first g, then f.
- ☐ *c* "Commutative" means $g \circ f = f \circ g$.
- ☐ *d* Commutative means: any two directed paths with the same start and end give the same map.

16 Which are abelian groups *(abelsche Gruppen)*?

- ☐ *a* $(\mathbb{Z}, +)$
- ☐ *b* $(\mathbb{Z}, \cdot)$
- ☐ *c* $(\mathbb{R}, \cdot)$
- ☐ *d* $(\mathbb{C}, +)$

17 Planes in $\mathbb{R}^3$.

- ☐ *a* Any three points determine exactly one plane.
- ☐ *b* Three non-collinear points p, q, r determine exactly one plane, $\{p + s(q-p) + t(r-p) \mid s, t \in \mathbb{R}\}$.
- ☐ *c* A plane through the origin is $\operatorname{span}(u, v)$ for linearly independent u, v.
- ☐ *d* $p + \operatorname{span}(u, v)$ is a plane whenever $u \neq 0$ and $v \neq 0$.

18 Let $z = a + bi$ with $a, b \in \mathbb{R}$, and $\bar{z} = a - bi$.

- ☐ *a* $z\bar{z} = a^2 - b^2$
- ☐ *b* $1/i = i$
- ☐ *c* $|z| = a + b$
- ☐ *d* $i^3 = i$

19 Which maps are surjective but not injective?

☐ *a* $\mathbb{R} \to [0, \infty)$, $x \mapsto x^2$ ☐ *b* $\mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$

☐ *c* $[0, \infty) \to [0, \infty)$, $x \mapsto x^2$ ☐ *d* $\mathbb{R} \to [-1, 1]$, $x \mapsto \sin x$

20 Linear combinations *(Linearkombinationen)* and the span *(lineare Hülle)*.

☐ *a* A linear combination of $v_1, \dots, v_k$ is $\lambda_1 v_1 + \cdots + \lambda_k v_k$ with $\lambda_i \in \mathbb{R}$.

☐ *b* $\operatorname{span}(v_1, \dots, v_k)$ is the set of all linear combinations of $v_1, \dots, v_k$.

☐ *c* 0 is a linear combination of any $v_1, \dots, v_k$.

☐ *d* $(3, 0) \in \operatorname{span}\big((1, 0), (2, 0)\big)$.

21 Rings and fields *(Ringe und Körper)*.

☐ *a* Every field is a ring.

☐ *b* Every ring is a field.

☐ *c* $(\mathbb{Z}, +, \cdot)$ is a ring but not a field.

☐ *d* $\mathbb{Z}_2 = \{0, 1\}$, with $+$ and $\cdot$ modulo 2, is a field.

22 Let $a \in \mathbb{R}^n$ with $a \neq 0$, let $b \in \mathbb{R}$, and $H = \{x \in \mathbb{R}^n \mid a \cdot x = b\}$.

☐ *a* H is a hyperplane.

☐ *b* H is a subspace *(Unterraum)* of $\mathbb{R}^n$, for every b.

☐ *c* $H = p + \operatorname{span}(v_1, \dots, v_{n-1})$ for some $p \in H$ and some linearly independent $v_1, \dots, v_{n-1}$.

☐ *d* If $v_1, \dots, v_{n-1}$ are linearly dependent, $p + \operatorname{span}(v_1, \dots, v_{n-1})$ still has dimension $n - 1$.

23 The notation $f \colon M \to N, m \mapsto f(m)$.

☐ *a* M is the domain *(Definitionsmenge)*, N the codomain *(Zielmenge)*.

☐ *b* Every $m \in M$ is assigned exactly one element $f(m) \in N$.

☐ *c* Every $n \in N$ equals $f(m)$ for some $m \in M$.

☐ *d* $\to$ goes between sets, $\mapsto$ between elements.

24 Linear independence in $\mathbb{R}^n$.

☐ *a* Every set that contains the zero vector is linearly dependent.

☐ *b* Every subset of a linearly independent set is linearly independent.

☐ *c* Every subset of a linearly dependent set is linearly dependent.

☐ *d* A linearly independent subset of $\mathbb{R}^n$ has at most n elements.

25 Image and codomain.

☐ *a* For every map $f \colon M \to N$: $f(M) \subseteq N$.

☐ *b* For $f \colon \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$, the image is a proper subset *(echte Teilmenge)* of the codomain.

☐ *c* If the image of f is a proper subset of its codomain, f is not surjective.

☐ *d* If the image of f is a proper subset of its codomain, f is not injective.

PART II

The Traps

Every distractor on a multiple-choice sheet is a true statement with one thing broken. Here is what breaks, by topic. The pool numbers point into the Musterantworten.

I have read every exam since there were examiners. The distractors come in four kinds: a reversed arrow, a dropped absolute value, a forgotten zero, a rule without its domain.

MAPS // ABBILDUNGEN · POOL 1–8

- **The rule is not the map.** Domain and codomain decide. $x \mapsto x^2$ is neither injective nor surjective on $\mathbb{R} \to \mathbb{R}$, surjective on $\mathbb{R} \to [0, \infty)$, bijective on $[0, \infty) \to [0, \infty)$.
- **Injective runs backwards:** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. The forward direction holds for every map, which makes it the favourite distractor.
- **Image ⊆ codomain, always;** equal exactly when f is surjective. A proper image says nothing about injectivity: e^x on $\mathbb{R} \to \mathbb{R}$. *Wertebereich* means the codomain in the Musterantworten, the image in some books. Read which one the question means.
- $g \circ f$ **means** f **first.** A diagram commutes when all directed paths with the same start and end give the same map. It has nothing to do with $g \circ f = f \circ g$.
- **"Or" is inclusive:** $x \in A \cup B$ allows x in both.

ALGEBRA // GROUPS, RINGS, FIELDS · POOL 9–12, 19

- **A group has three axioms:** associativity, a neutral element, inverses. Closure is built into $*\colon G \times G \to G$. Commutativity is extra; with it the group is abelian.
- **The usual non-groups:** $(\mathbb{N}, +)$ (no -1), $(\mathbb{Z}, \cdot)$ (no $\frac{1}{2}$), $(\mathbb{R}, \cdot)$ (0 has no inverse). Remove the zero and $(\mathbb{R} \setminus \{0\}, \cdot)$ is a group.
- **A field inverts everything except** 0, and has $0 \neq 1$. Fields: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_2$. Not a field: $\mathbb{Z}$.
- **Field ⇒ ring.** Never the converse.
- **In** $\mathbb{C}$, $i^2 = -1$. Multiply out, then replace i^2 by -1. To divide, multiply top and bottom by the conjugate: $\frac{z}{w} = \frac{z\bar{w}}{|w|^2}$ with $|w|^2 = w\bar{w} = c^2 + d^2$ for $w = c + di$. So $1/i = -i$ and $i^3 = -i$.

VECTOR SPACES // VEKTORRÄUME · POOL 25

- **Check the zero vector first.** No zero, no vector space: $\{x + y = 1\}$, and the polynomials of degree exactly 2 (also not closed: $x^2 + (-x^2) = 0$).
- **Examples to name:** $\mathbb{R}^n$, $P_2(\mathbb{R})$, all functions $\mathbb{R} \to \mathbb{R}$, $\mathbb{C}$ over $\mathbb{R}$, $\{0\}$.

INDEPENDENCE // LINEARE UNABHÄNGIGKEIT · POOL 13–14, 18, 20–22, 26–27

- **The word is "only":** $\sum \lambda_i v_i = 0$ *only* for all $\lambda_i = 0$. That zero coefficients give 0 is true for every family of vectors.
- **The zero vector makes any set dependent:** $1 \cdot 0 = 0$ is a nontrivial combination. All vectors nonzero is still not enough: $(1, 0), (2, 0)$.
- **Two vectors are dependent** exactly when one of them is a multiple of the other. "One of them" matters: 0 is a multiple of every v, but $v \neq 0$ is no multiple of 0.
- **Count:** at most n independent vectors in $\mathbb{R}^n$. Two numbers in $\mathbb{R}$ are always dependent; three vectors in $\mathbb{R}^2$ too.
- **Inheritance:** subsets of independent sets are independent, supersets of dependent sets are dependent. Not the other way round.
- **Linear combinations are finite sums;** the span is the set of all of them, and 0 is always in it.

GEOMETRY // GERADEN, EBENEN, HYPEREBENEN, NORM · POOL 15–17, 23–24

- **Line:** two *distinct* points, or a point and a direction $\neq 0$. Through the origin: one vector $\neq 0$. Parallel direction vectors give the same line.
- **Plane:** three *non-collinear* points, or a point and two *independent* directions. Through the origin: two independent vectors.
- **Hyperplane: dimension** $n-1$**. Plane: dimension** 2**.** The two agree only in $\mathbb{R}^3$. In $\mathbb{R}^2$ a hyperplane is a line; in $\mathbb{R}^4$ it is 3-dimensional.
- $\{a \cdot x = b\}$ **with** $a \neq 0$ is a hyperplane, and a subspace only when $b = 0$. Written as $p + \operatorname{span}(v_1, \dots, v_{n-1})$, the v_i must be independent, or the span has fewer than $n-1$ dimensions.
- **Norm:** $\|x\|_2 = \sqrt{x_1^2 + \cdots + x_n^2} \geq 0$, $\|x\| = 0 \iff x = 0$, $\|\lambda x\| = |\lambda| \, \|x\|$ with the absolute value, and $\|x + y\| \leq \|x\| + \|y\|$.

PART III

The Key

Bold: the true boxes. Then why each false box is false.

01 **b** POOL 3 (a) $(-1)^2 + 1 = 2$: the square eats the sign. (c) 0 is never reached. (d) $f(1) = f(-1) = 2$.

02 **a, d** POOL 13, 27 (b) True for every family; the definition says *only*. (c) Necessary, not sufficient: $(1, 0), (2, 0)$.

03 **a, d** POOL 9 (b) That is the extra axiom of an abelian group. (c) 1 has no inverse: $1 + n = 0$ has no solution in $\mathbb{N}$.

04 **a, c** POOL 15 (b) $p = q$ gives no direction. (d) $(2, 4) = 2(1, 2)$: same direction through the same point, same line.

05 **a, b** POOL 1 (c) "Or" is inclusive: 3 is in both. (d) $A \cap \emptyset = \emptyset$.

06 **a, c** POOL 19 (b) $(1+2i)(1-i) = 1 - i + 2i - 2i^2 = 3 + i$; $-1 + i$ comes from $i^2 = +1$. (d) $i^2 = -1$. Check (c): $(1-i) \cdot \frac{-1+3i}{2} = \frac{2+4i}{2} = 1 + 2i$.

07 **a, b** POOL 25 (c) $x^2 + (-x^2) = 0$ has degree $\neq 2$: not closed, no zero vector. (d) $(0,0) \notin$ the set.

08 **a, c** POOL 5 (b) Same rule, smaller codomain: now bijective. (c) The smallest natural number (0 or 1, by your convention) is never hit. (d) Bijective, with inverse $\sqrt[3]{x}$.

09 **a, b, c** POOL 14, 18, 21, 22 (a) $2(1,2) - (2,4) = 0$. (b) $2 \cdot 1 - 1 \cdot 2 = 0$. (c) $2e_1 - 1 \cdot (2e_1) = 0$. (d) $\lambda + \mu = 0$ and $\lambda - \mu = 0$ force $\lambda = \mu = 0$.

10 **b, c, d** POOL 11 (a) Every element *except* 0.

11 **a, b, c** POOL 17 (d) A plane in $\mathbb{R}^4$ has dimension 2, a hyperplane 3.

12 **a, c, d** POOL 4 (b) Holds for every map; the arrow points the wrong way.

13 **a, b, c** POOL 24 (d) $\|\lambda x\| = |\lambda| \, \|x\|$. With $\lambda = -1$ the right side would be negative.

14 **b, d** POOL 14 (a) $\dim \mathbb{R} = 1$. (c) At most 2 independent vectors in $\mathbb{R}^2$.

15 **a, d** POOL 8 (b) $(g \circ f)(a) = g(f(a))$: f acts first. (c) $f \circ g$ is not even defined here: g lands in C, f starts in A.

16 **a, d** POOL 10 (b) $2x = 1$ has no solution in $\mathbb{Z}$. (c) 0 has no inverse.

17 **b, c** POOL 16 (a) Three collinear points lie on infinitely many planes. (d) $u = (1,0,0)$, $v = (2,0,0)$ span a line.

18 **none** POOL 19 (a) $z\bar{z} = a^2 + b^2$. (b) $i \cdot (-i) = 1$, so $1/i = -i$. (c) $|z| = \sqrt{a^2 + b^2}$. (d) $i^3 = i^2 \cdot i = -i$.

19 **a, d** POOL 6 (b) Neither: -1 is never hit, and $f(1) = f(-1)$. (c) Bijective, with inverse $\sqrt{x}$.

20 **a, b, c, d** POOL 26 (c) Take every $\lambda_i = 0$. (d) $(3,0) = 3(1,0) + 0 \cdot (2,0)$.

21 **a, c, d** POOL 12 (b) $\mathbb{Z}$ is a ring and not a field. (d) The only nonzero element, 1, is its own inverse.

22 **a, c** POOL 17, 23 (b) For $b \neq 0$, $0 \notin H$. (d) Dependent vectors span fewer than $n - 1$ dimensions; that is why independence appears in the description.

23 **a, b, d** POOL 2 (c) That is surjectivity, not part of being a map.

24 **a, b, d** POOL 13, 27 (c) $\{(1,0)\} \subset \{(1,0), (2,0)\}$: dependent set, independent subset.

25 **a, b, c** POOL 7 (d) e^x on $\mathbb{R} \to \mathbb{R}$: injective, image $(0, \infty)$.